

1. Summarising Notes

Prompt

Make the following notes brief and clear with bullets for exam preparation:

“Data Structures: Stack works on LIFO rule. Queue works on FIFO. Tree is a hierarchy. Graph has nodes and edges. Searching techniques are linear and binary search.”

AI Response

- >LIFO is used in Stack
- >FIFO is used in Queue
- >Tree is a hierarchical structure
- >Graph is made up of nodes (vertices) and edges
- >Linear search and binary search are used in searching techniques

How this prompt helps

It explicitly states:

Task is mentioned that is summarising.

Output is given with bullets

Complexity level is simple and understandable

This ensures that AI gives concise answers.

2. Generating a Study Plan

Prompt

“Create a 5-day study plan for Data Structures exam preparation. Include daily topics, revision time, and practice questions. Keep it balanced and realistic for a college student.”

AI Response

Day 1: Arrays & Strings

- Study concepts (2 hrs)
- Practice problems (1 hr)
- Quick revision (30 min)

Day 2: Stack & Queue

- Learn implementation (2 hrs)
- Solve problems (1 hr)

Day 3: Linked List

Types and operations (2 hrs)

- Coding practice (1 hr)

Day 4: Trees

- Traversals and BST (2 hrs)
- Problem solving (1 hr)

Day 5: Graphs + Revision

- Graph basics (2 hrs)
- Full revision + mock test (2 hrs)

Why this prompt works

It gives:

- Clear duration (5 days)
- Required structure (daily plan + revision + practice)
- Target audience (college student)
So AI generates organized scheduling instead of random topics.

3. Creating Interview Questions

Prompt

“Generate 8 technical interview questions on Python basics and Data Structures for a beginner-level job interview. Include a mix of theory and coding questions.”

☐ **AI Response**

1. What is the difference between list and tuple in Python?
2. Explain what a stack is with an example.
3. What is a queue and where is it used?
4. Write a Python program to reverse a string.
5. What are Python dictionaries?
6. Explain time complexity in simple terms.
7. Write code to implement a stack using a list.
8. What is a linked list?

☐ **Why this prompt works**

It defines:

- Number of questions (8)
- Difficulty level (beginner)
- Topics (Python + DS)
- Type (theory + coding mix)
This ensures balanced and relevant output.

4. Writing a Resume Summary

☐ Prompt

“Write a professional resume summary for a 2nd-year Computer Science student who knows Python, Java, and basic web development. Keep it 3–4 lines and job-ready.”

☐ AI Response

Motivated Computer Science student with foundational knowledge in Python, Java, and web development technologies. Strong interest in software development and problem-solving. Familiar with data structures and algorithms. Seeking opportunities to apply technical skills in real-world projects and internships.

☐ Why this prompt works

It specifies:

- Role (2nd-year CSE student)
 - Skills (Python, Java, web dev)
 - Length (3–4 lines)
 - Tone (professional)
- This ensures a polished and job-ready summary.

5. Creating a Lesson Plan

☐ Prompt

“Create a lesson plan to teach beginners the concept of queues in Data Structures. Include explanation, real-life example, and a small coding activity.”

☐ AI Response

Topic: Queue in Data Structures

Explanation:

A queue is a linear data structure that follows FIFO (First In First Out), meaning the first element added is removed first.

Real-life Example:

People standing in a ticket line—first person in line gets served first.

Coding Activity:

Write a Python program to implement a queue using a list:

- Enqueue (add element)

- Dequeue (remove element)
- Display queue

□ **Why this prompt works**

It clearly includes:

- Audience (beginners)
 - Required sections (explanation, example, coding task)
 - Topic focus (queues)
- This guides AI to structure teaching content properly.